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System Analysis and Design (SECD2613)

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1.0 Overview of the Project

In this report, we will outline the development and design of a proposed system aimed at addressing inefficiencies in the current business process. We begin this project with a detailed analysis of the existing workflow, identifies the key challenges and proposes optimized solutions through systematic redesign.

The report follows a structured approach, starting with by stating the problem statement to highlight the issues faced , followed by an analysis of the current business process/workflow using AS-IS logical DFD. The proposed solutions are then presented through a TO-BE logical DFD which includes context diagrams, process specification and refined workflows.

Proceed to the next section, the physical system design section will translate the logical model into a technical blueprint. The system wireframe section provides visual representation of input and output designs to enhance user interaction.

This report ends with a summary of the proposed system, to emphasize the benefits and expected impact on operational efficiency. This comprehensive documentation ensures clarity in system requirements, design and implementation which will serve as a foundation for future development.

2.0 Problem Statement

Through interviews and surveys conducted with UTM students and UTM Fleet staff, the current system faces several significant issues. These problem limit proper shuttle operation and affect students transportation routine. The key issues faced by users and stakeholders are outlined below:

1. Lack of Real-Time Shuttle Tracking

The absence of GPS tracking causes the students to have no way to know the exact location of shuttles and when they will arrive at a specific stop. This can cause students to sometimes miss the shuttle and arrive late to class. Having real-time bus information is compulsory as being on time is important especially to students who stay far from their faculty. UTM Fleet management team also have problems in tracking their own buses location in case there are delays or changes when there are problems. By providing real-time updates and important data to enhance the shuttle service's operation, a real-time tracking system would benefit both shuttle staff and students.

2. Shuttle Information is Static

Currently students can find shuttle bus schedules through UTM website online by scanning the qr code that will be directing students to the bus schedules. Students can download the schedules and view the route details at any time. However, the shuttle information is static as a pdf document and might be confusing to new students or those who are not familiar with the routes. Students must scroll through the entire document just to find simple details like which bus stop to go to or when the next shuttle is expected. The document also does not offer interactive features such as a search bar, filters, or clickable maps which is not practical. A more user-friendly platform, such as a mobile app with search and filter functions, would make it much easier for students to check shuttle times and plan their journey on campus.

3. No Notification or Alert System for Shuttle Updates

In the absence of a notification or alert system when there are delays, maintenance issues, or sudden changes in the shuttle schedule causing students to not be informed quickly or directly. Most of the time, these updates are either not shared at all or only posted on social media platforms, which not all students follow and alert. As a result, many students only find out that the shuttle is late or not coming after they have waited at the stop. This lack of communication causes inconvenience and wastes students' time, especially during busy hours or bad weather. Students are unable to make quick decisions, such as finding another route or leaving earlier. A proper alert system, such as push notifications through a mobile app would help students stay informed about any changes in the shuttle service.

4. Lack of a feedback mechanism for sharing ideas or reporting issues

At the moment, students do not have an official way to report a problem with the shuttle. There is no mobile app or platform where they can report any problem without feel shy or uncomfortable making complaints by phone call directly to UTM Fleet hotline number, especially for small or repeated problems like overcrowded buses or unclear shuttle signs. As a result, many issues are not reported, and the same problems continue to happen every day. It is difficult for the UTM Fleet management team to improve their shuttle system because they may not be informed about the problems students are actually facing. A better solution would be to include a feedback form or chatbot in a mobile app. This would allow students to report problems anytime and help UTM Fleet improve their services based on real student experiences. It would also make students feel heard and involved in improving campus.

5. Lack of Reporting or Data Gathering Tools for Service Enhancement

UTM Fleet does not have a proper system to collect and analyze data from shuttle users. There is no record regarding the number of students using each shuttle, the busiest hours, or the most frequently used routes. Without this information, they cannot make decisions on what specification they should improve in their service. For example, they do not know if they need to add more buses during peak hours or adjust the schedule based on actual usage patterns. In addition, there is no performance tracking system to measure how often buses arrive late, how many complaints are received, or how satisfied students are with the service. All of these are important for managing and improving a transportation system. A smart shuttle system that collects real-time data and creates simple reports would help UTM Fleet understand what is working and what needs to be fixed.

3.0 Proposed Solutions

UTM SmartShuttle is the new system solution that we propose to introduce to the stakeholder, the UTM Fleet Management, with the main objective of bringing convenience, efficiency, and real-time communication to UTM' shuttle transportation system. This proposed system aims to revolutionize the way shuttle operations are monitored and experienced by both students and staff. It focuses on solving all the key issues highlighted earlier through an integrated mobile application platform.

This system will provide real-time shuttle tracking using GPS technology. Students will be able to see the exact location of every shuttle, the estimated time of arrival of each stop, and route directions through an interactive map. By knowing when and where the shuttle will arrive, students can plan their journey better and avoid waiting long under uncertain conditions. UTM Fleet staff can also monitor all shuttle movements through a backend admin dashboard, allowing them to respond to route delays, reroute in emergencies, and optimize service schedules.

In addition to live tracking, the app will replace the static PDF shuttle schedules currently available with a more user-friendly interface. The updated platform will include interactive features such as a searchable timetable, filter options by location or shuttle route, and a clickable route map to help students quickly find the nearest shuttle stop and understand route details. This eliminates confusion for new students or those unfamiliar with the shuttle system, improving overall navigation across campus.

Furthermore, a notification and alert system will be integrated to send important updates instantly. Students will receive push notifications about any changes to the shuttle service, such as delays, route changes, maintenance issues, or emergency alerts. These alerts will be sent through the app and optionally via email or SMS. This ensures students are well-informed without having to rely on social media platforms, which not all students may follow actively.

The system will also introduce a feedback and reporting mechanism that allows students to easily submit suggestions, complaints, or incident reports via the app. Whether it's about overcrowded shuttles, unclear signage, or service dissatisfaction, students can express their concerns anonymously and at any time. This feedback will be collected and categorized automatically, helping UTM Fleet identify recurring issues and make informed service improvements. The system may also include chatbot assistance, Aira, for faster response and guidance.

To support continuous service enhancement, the UTM SmartShuttle system will be equipped with data collect and reporting tools. These tools will track metrics such as passenger counts, shuttle punctuality, popular routes, peak hours, and common complaints. UTM Fleet managers can view visual reports, export usage summaries, and make data-driven decisions

to optimize shuttle frequency, adjust schedules, or allocate more buses to busy routes. This also helps in resource planning and performance evaluation of the shuttle system.

Moreover, UTM SmartShuttle will feature personalization options for users. Students can bookmark their most-used routes or stops, set reminder alerts for bus arrival times, and filter notifications based on their preferences. The backend system allows fleet administrators to update route info, schedules, and notification messages directly, which will be synced in real time to all users. A secure login and account system will ensure privacy and data protection for all users.

Lastly, the system will be designed with a modern, intuitive interface to ensure it is mobile-friendly. Students can access shuttle information anytime, anywhere, and receive timely updates that keep them informed, reduce waiting times, and improve their daily commuting experience on campus. For UTM Fleet, this system will enhance efficiency, transparency, and communication, leading to a more responsive and reliable shuttle service.

In summary, UTM SmartShuttle is not just a tracking app but it is a complete smart transportation solution that combines live data, communication, analytics, and user engagement to create a smoother and more connected commuting experience for the UTM community.

4.0 Current Business Process/Workflow

Simple explanation of the AS-IS system workflow focusing specifically on how students currently ride the UTM campus shuttle, step by step, from checking the schedule to boarding the bus.

1.0 Find the Shuttle Schedule

- 1.1 Students scan QR code on websites or at bus stops to get current shuttle schedules.
- 1.2 Download PDF schedule (no real-time updates).

2.0 Check shuttle time and plan to go to the bus stop

- 2.1 Refer to the fixed schedule in PDF.
- 2.2 Estimate time to wait at the nearest stop.

3.0 Walk to the nearest shuttle stop

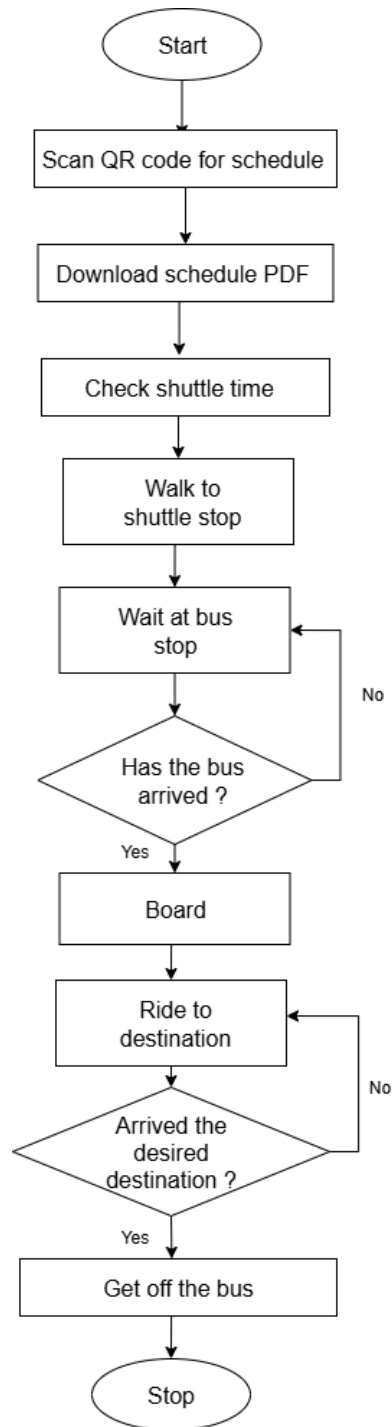
- 3.1 Student waits based on the printed schedule.

4.0 Board the shuttle

- 4.1 Student boards the shuttle when it arrives.

5.0 Ride the shuttle to destination

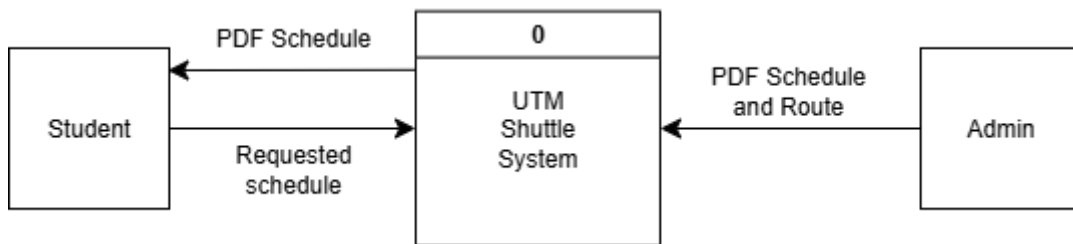
- 5.1 The driver follows a fixed route and schedule.
- 5.2 Students wait until their destination is reached.



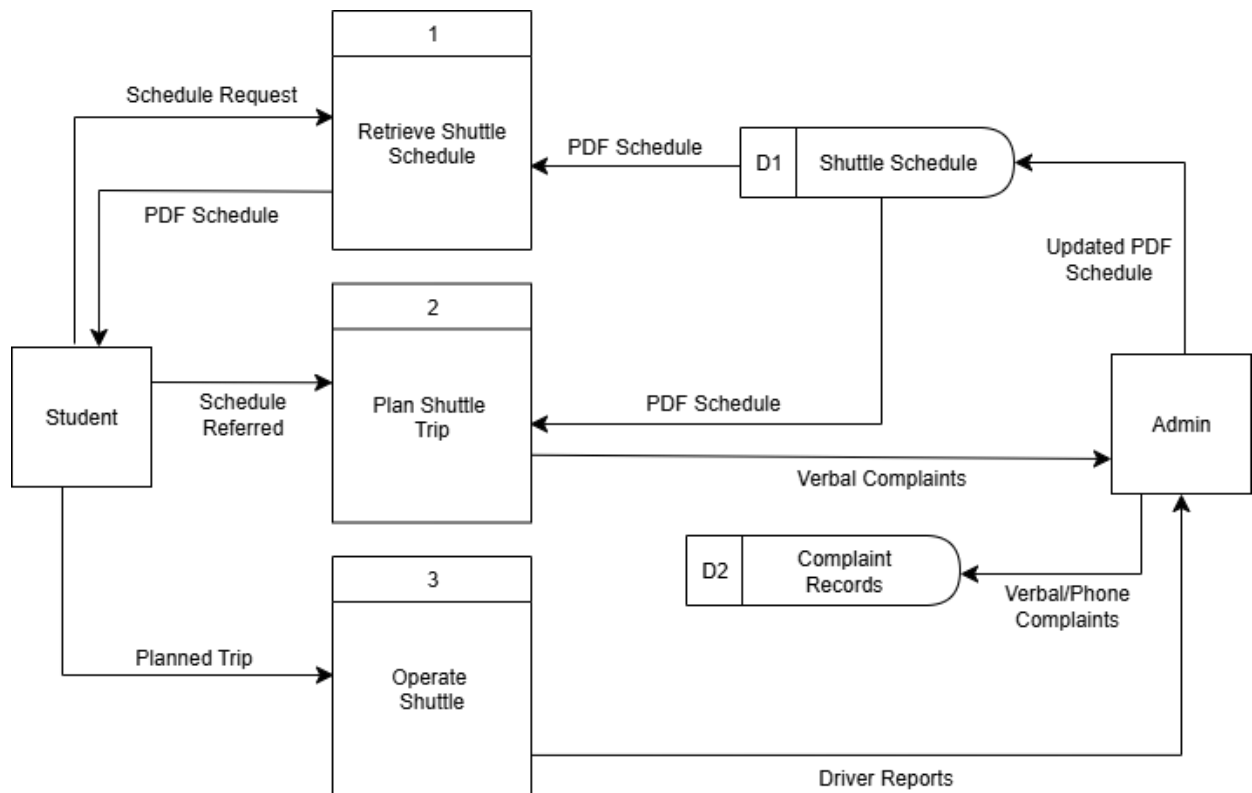
AS-IS System Workflow

5.0 Logical DFD (AS-IS)

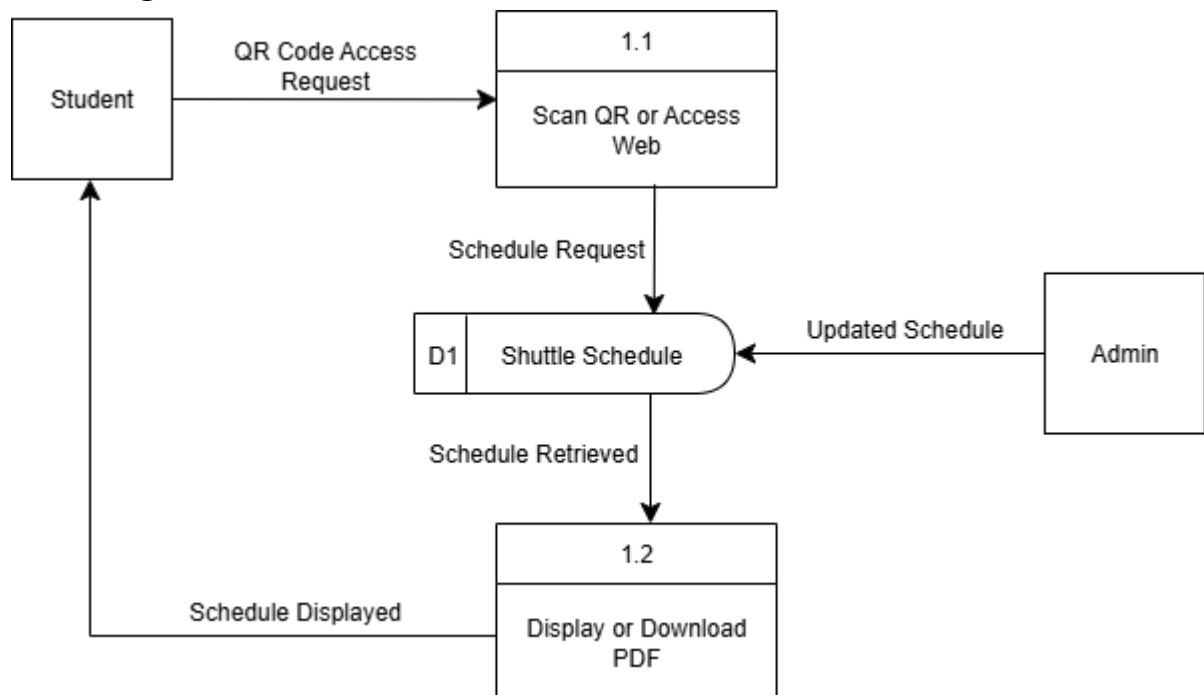
Context Diagram



Level 0 Diagram



Child Diagram



6.0 System Analysis and Specification

6.1 Logical DFD TO-BE system (Context Diagram, Diagram 0, Child)

Context Diagram

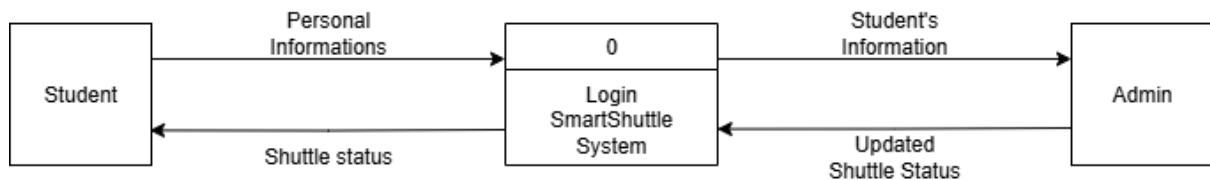
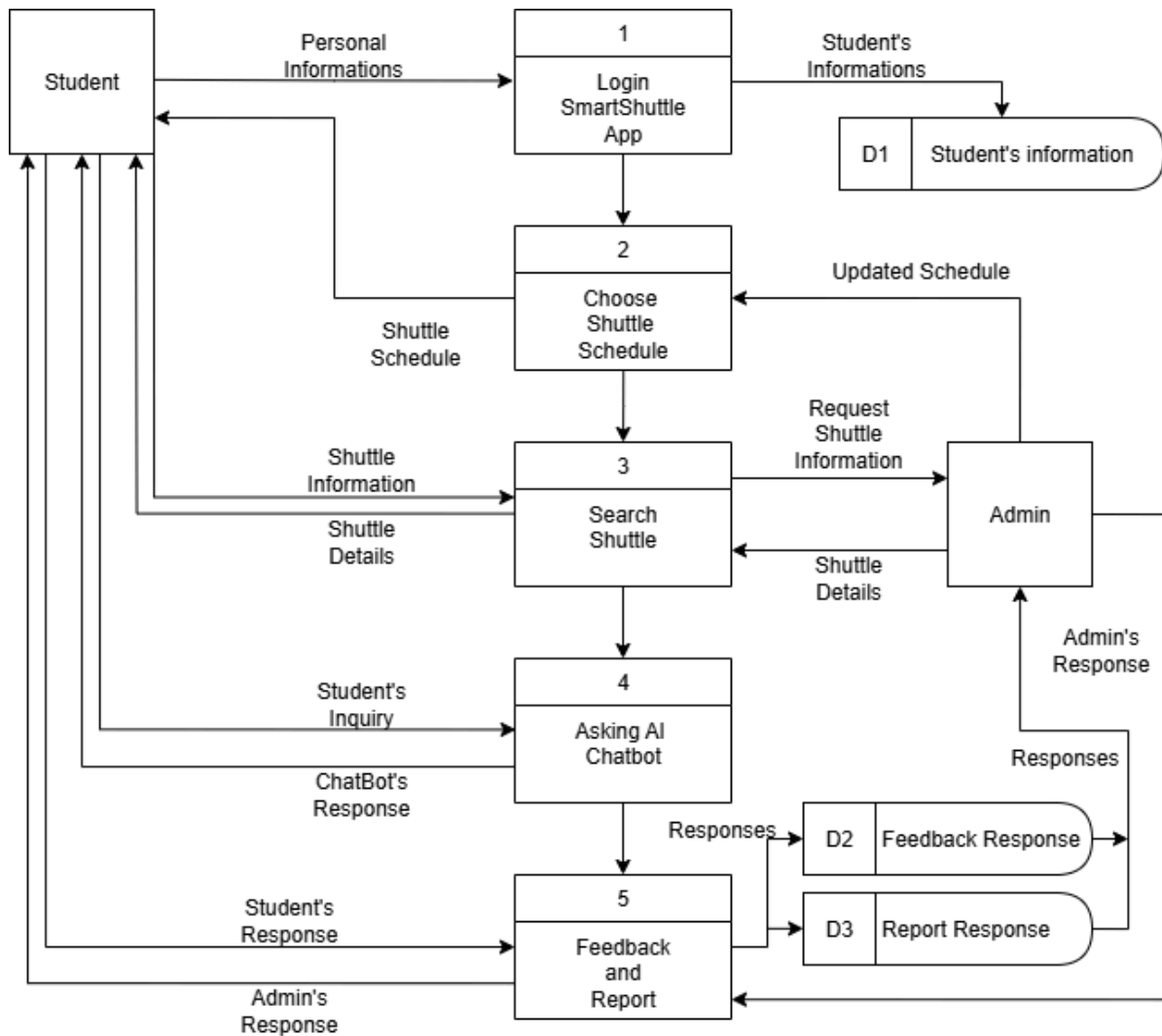
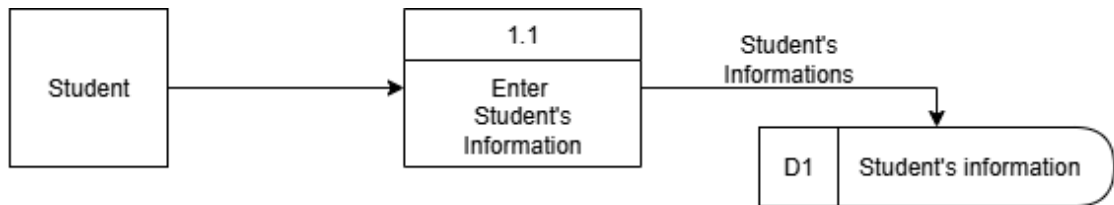


Diagram 0

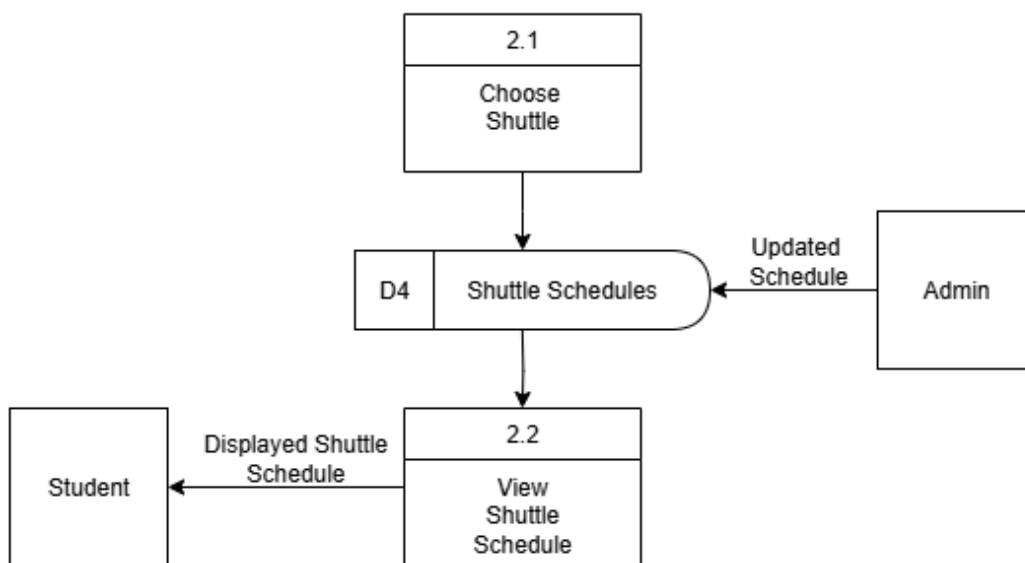


Child Diagram

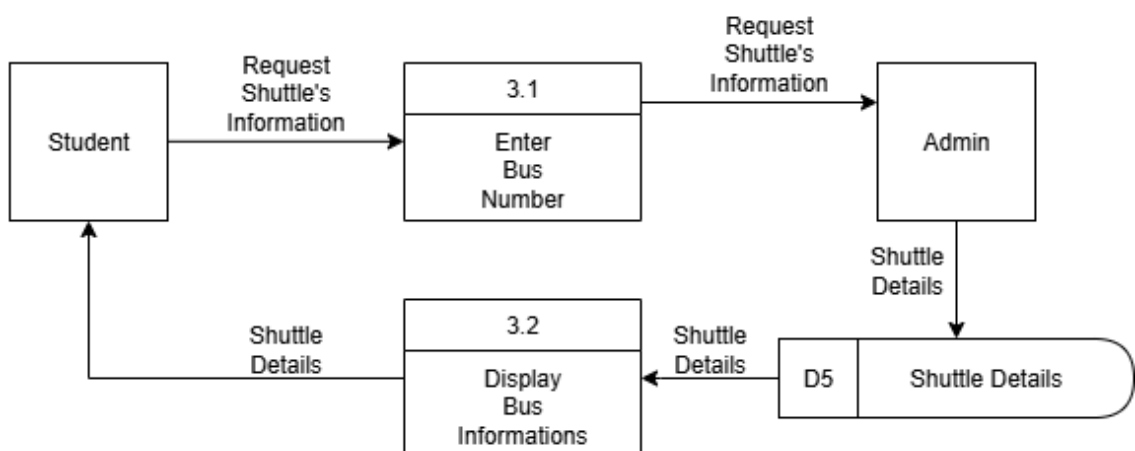
Process 1



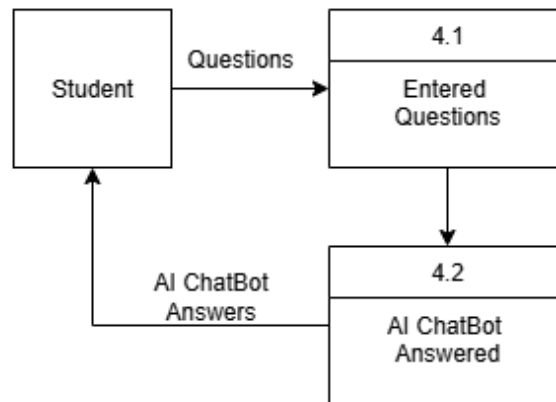
Process 2



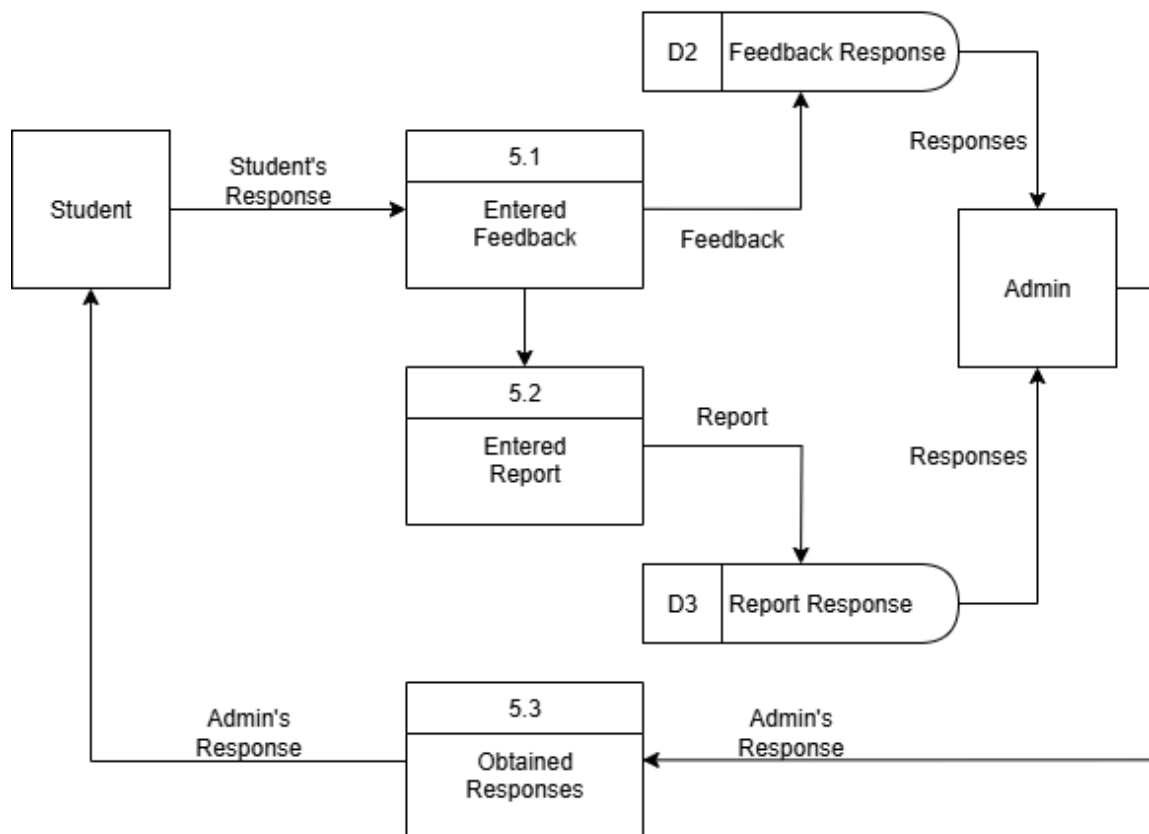
Process 3



Process 4



Process 5



6.2 Process Specification (based on Logical DFD TO-BE)

6.2.1 Process 1: Login SmartShuttle App

DO WHILE User Login

READ Personal Information from Student.

VALIDATE Personal Informations against existing records.

IF Personal Informations are VALID

CREATE/RETRIEVE Student's Information record.

STORE Student's Information in Student's Information.

GRANT access to the SmartShuttle App.

ELSE (Personal Informations are INVALID)

DISPLAY error message to Student.

PROMPT Student to re-enter Personal Informations

END IF

END DO

6.2.2 Process 2: Choose Shuttle Schedule

DO WHILE Admin update schedule.

RECEIVE Updated Schedule from Admin.

UPDATE Shuttle Schedules with Updated Schedule.

RETRIEVE current Shuttle Schedules from Shuttle Schedules.

RECEIVE Shuttle Schedule selection from Student.

CONFIRM Shuttle Schedule selection to Student

DISPLAY Shuttle Schedules to Student

END DO

6.2.3 Process 3: Search Shuttle

DO

PROMPT Student to ENTER Bus Number.

RECEIVE Bus Number from Student.

VALIDATE Bus Number format.

IF Bus Number is VALID

SEND Request Shuttle Information (containing Bus Number) to Admin.

ELSE (Bus Number is INVALID)

DISPLAY error message to Student.

PROMPT Student to re-enter Bus Number.

ENDIF

RECEIVE Shuttle Details from Admin.

STORE Shuttle Details in Shuttle Details.

RETRIEVE Shuttle Details from Shuttle Details.

FORMAT Shuttle Details for display.

DISPLAY Formatted Shuttle Details to Student

END DO

6.2.4 Process 4: Asking AI ChatBot

DO

RECEIVE Student's Inquiry from Student.

PROCESS Student's Inquiry using AI Chatbot logic.

GENERATE ChatBot's Response based on inquiry and available data.

DISPLAY ChatBot's Response to Student.

END DO

6.2.5 Process 5: Feedback and Response

DO

RECEIVE Student's Response from Student.

DETERMINE if Student's Response is Feedback or a Report.

IF Student's Response is Feedback:

STORE Feedback in Feedback Response.

SEND Feedback to Admin.

ELSE IF Student's Response is a Report:

STORE Report in Report Response.

SEND Report to Admin.

RECEIVE Responses from Admin.

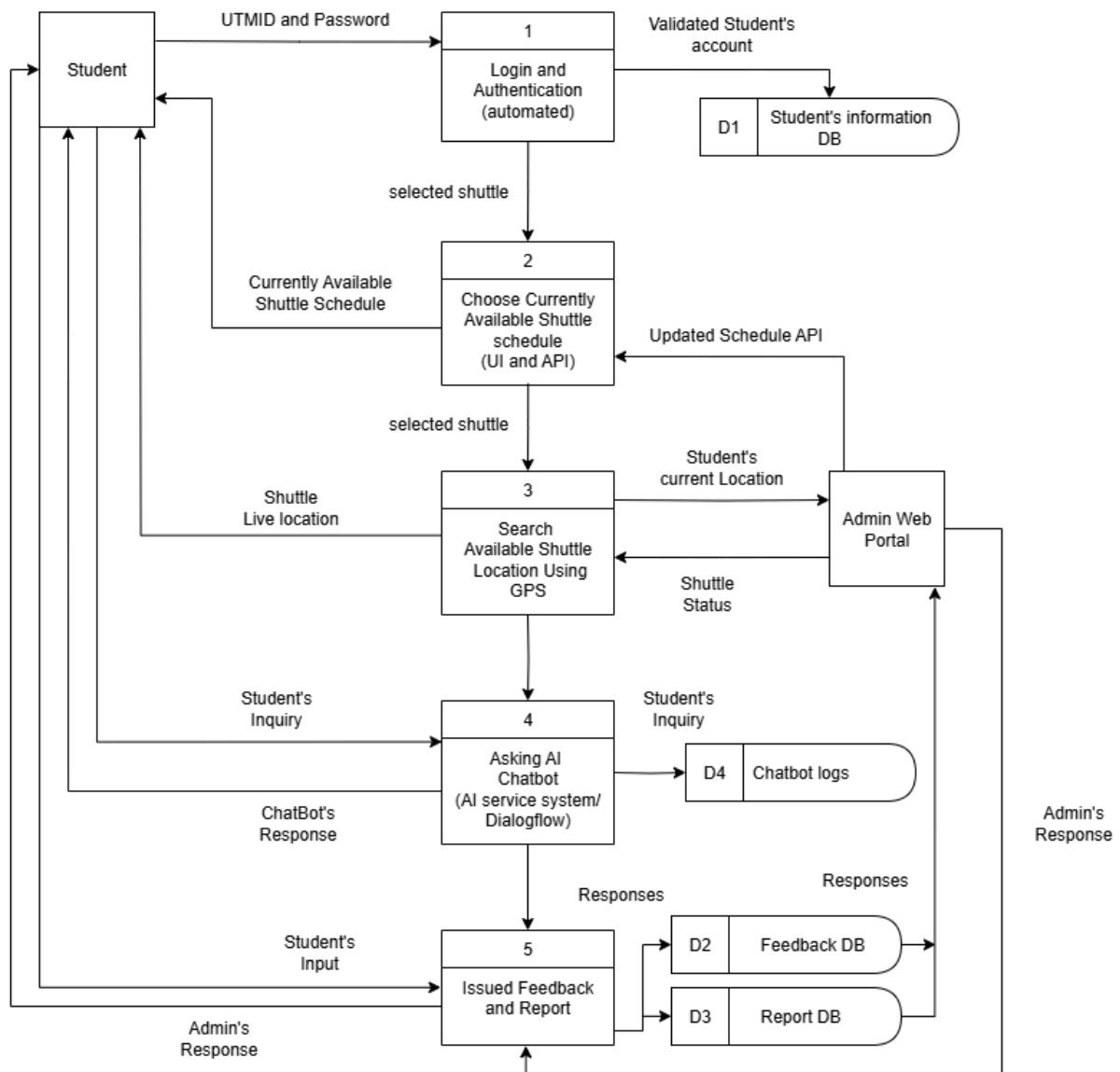
DISPLAY Admin's Response to Student.

END DO

7.0 Physical System Design

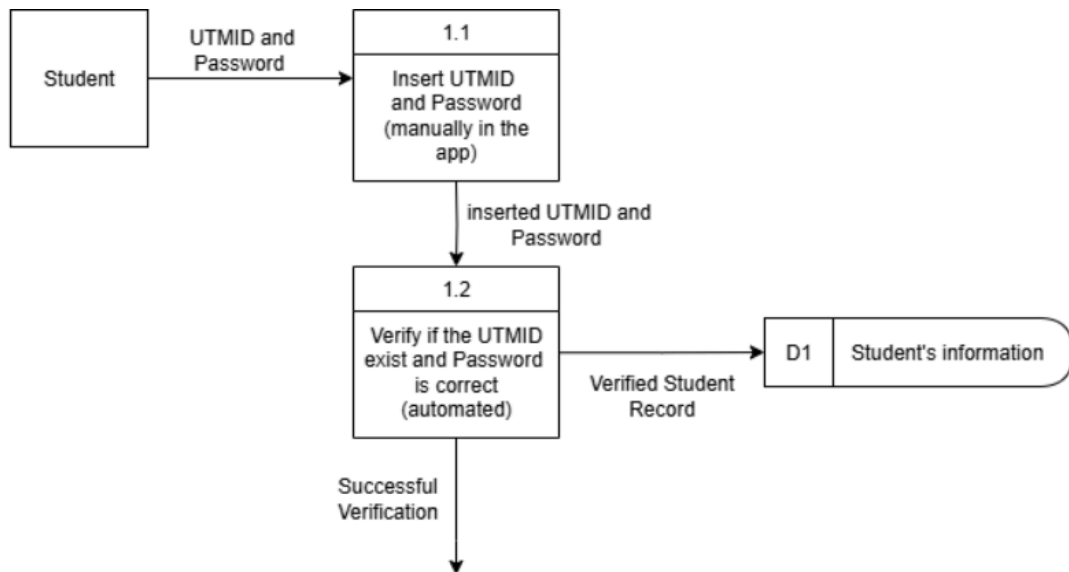
7.1 Physical DFD TO-BE system (Diagram 0, Child, Partitioning, CRUD Matrix)

7.1.1 0 Diagram

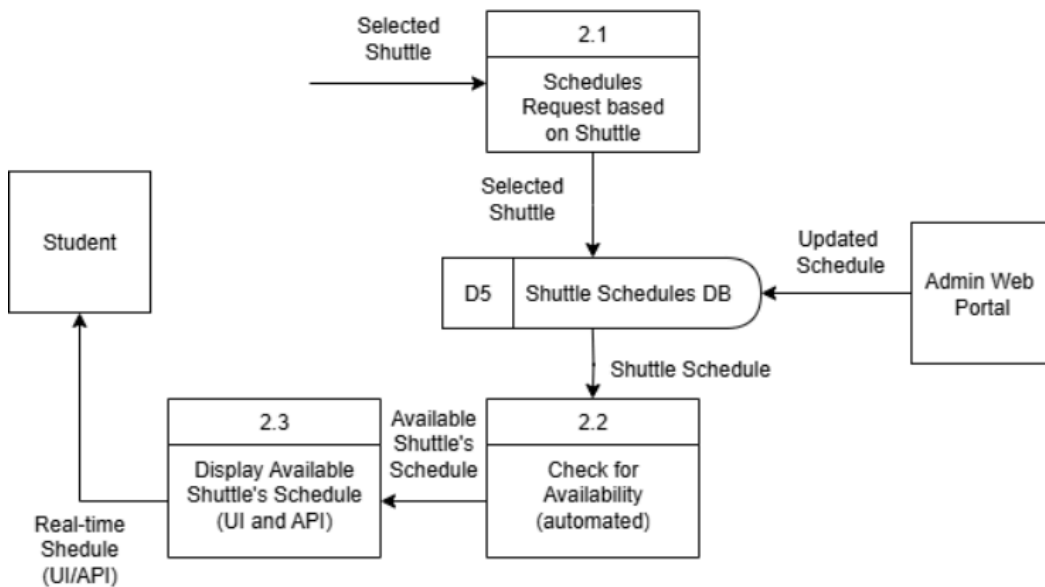


7.1.2 Child Diagram

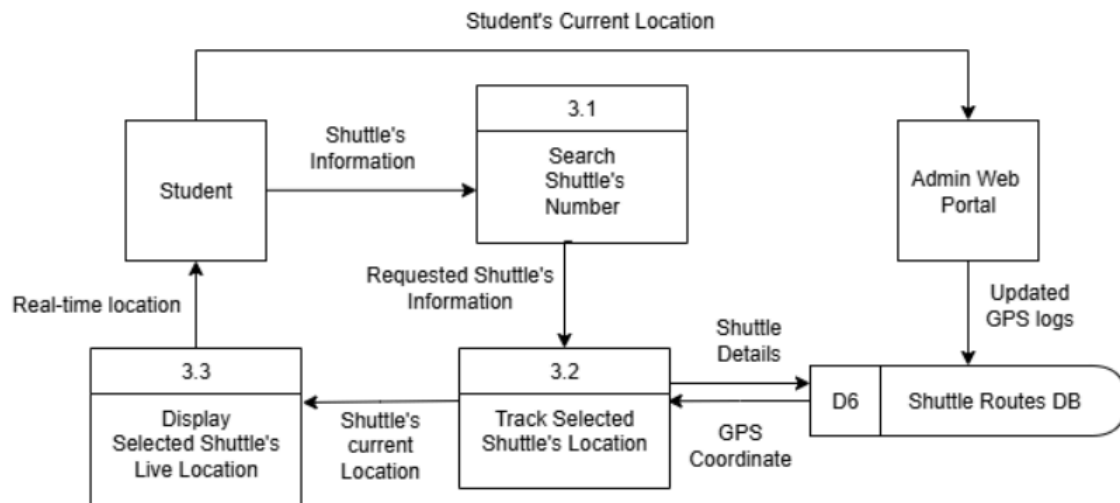
Process 1: Log In and Authentication



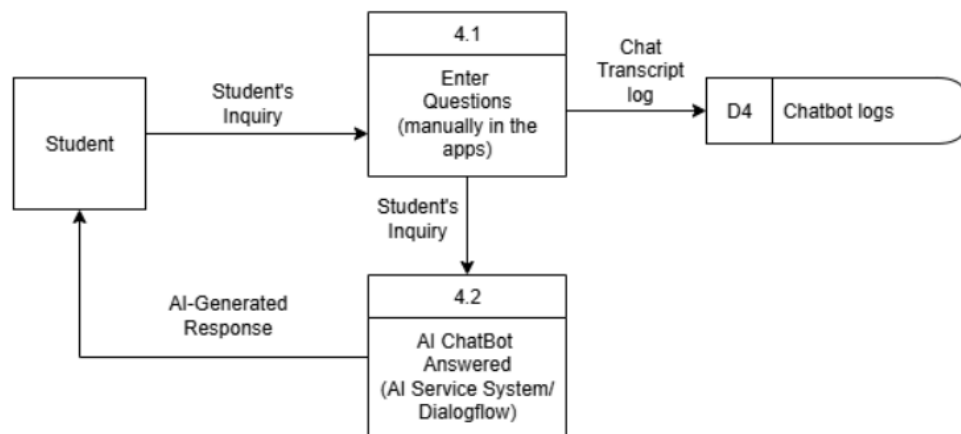
Process 2: Choose Available Shuttle Schedule



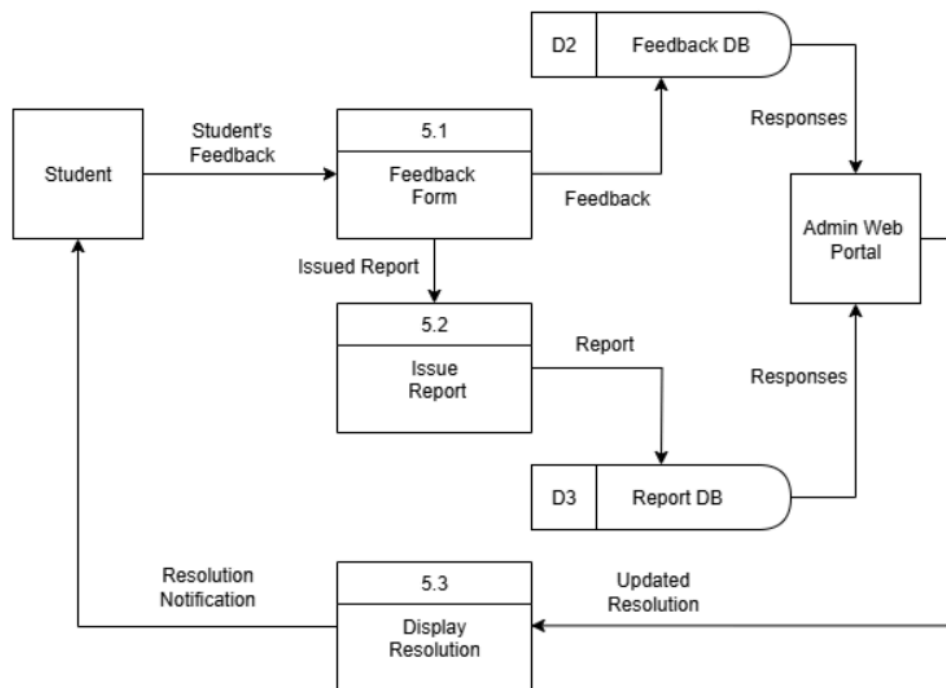
Process 3: Search Available Shuttle Location



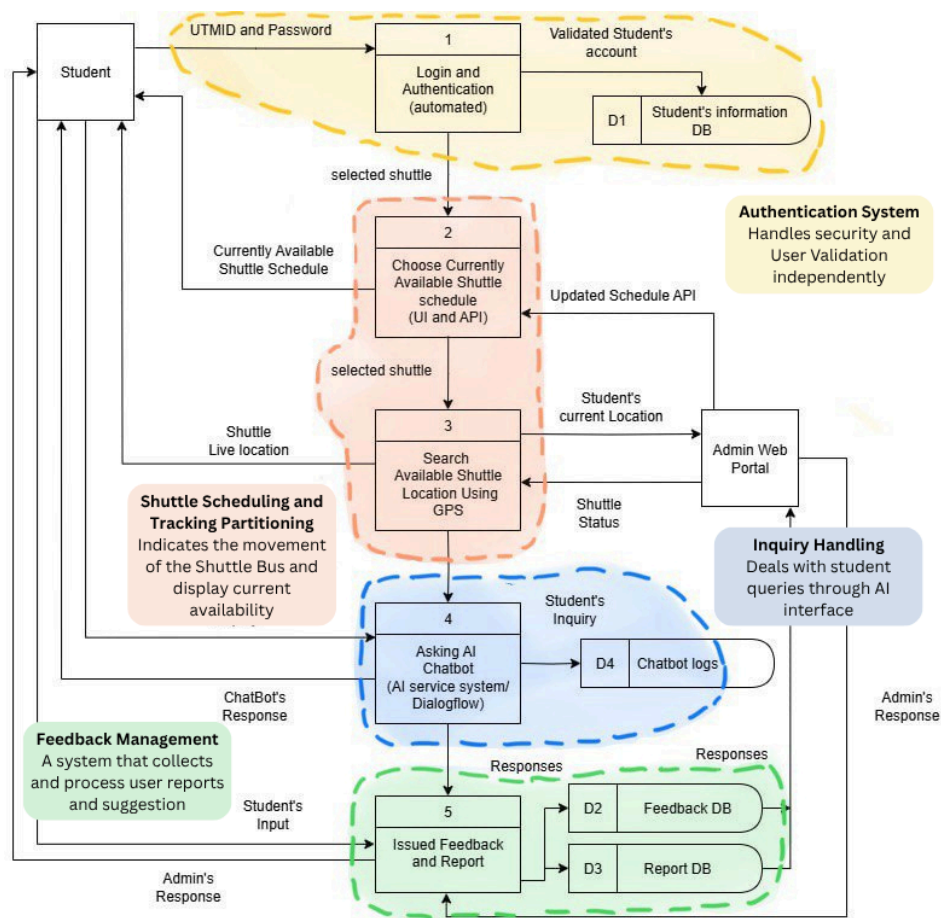
Process 4: Asking AI Chatbot



Process 5: Issue Feedback and Report



7.1.3 Partitioning



Authentication System Partitioning

This partitioning includes only Process 1 as it focuses exclusively on user identity verification and access control. Given that authentication handles sensitive information such as personal credentials, it is logically and physically separated to enforce strict security measures. Isolating this function ensures that security protocols can be specialized without affecting the performance of other components. Furthermore, this separation enhances system efficiency by ensuring that any potential authentication latency does not interfere with core operational tasks like shuttle schedule lookups, thereby optimizing overall system responsiveness.

Shuttle Scheduling and Tracking Partitioning

This partitioning includes Process 2 and Process 3, as both processes focus primarily on shuttle operations management. Grouping these functions together allows better control over shuttle-related tasks such as schedule selection, updating, and real-time shuttle tracking. Besides, the system can maintain operational efficiency, reduce dependencies, and improve scalability for shuttle services. This partitioning also allows shuttle-related modules to be updated or optimized independently, ensuring a more modular and maintainable architecture.

Inquiry Handling Partitioning

This partitioning only includes Process 4 which is related to Artificial Intelligence (AI) chatbot services. This will allow independent scaling for the chatbot as it might need to handle thousands of concurrent inquiries. Moreover, isolating this module supports continuous learning and improvement of the AI model through chatbot logs, contributing to more accurate and efficient responses over time.

Feedback Management Partitioning

This partitioning includes Process 5, which is dedicated to collecting, processing, and storing user feedback and incident reports. Isolating this function ensures that the data collection mechanisms and reporting logic do not interfere with core user interactions such as authentication, scheduling, or inquiries. By partitioning feedback management, the system supports data integrity, facilitates better issue tracking and service improvement, and allows for asynchronous processing of reports without impacting user-facing performance.

7.1.4 CRUD Matrix

Diagram 0

Activity	Student	Admin	Student's Information DB	Chatbot Logs	Feedback DB	Report DB
Login and Authentication	C		R			
Choose Shuttle Schedule	CR	RU				
Search Shuttle Location	CR	RU				
Asking AI Chatbot	C			R		
Issue Feedback and Report	C	RU			R	R

Process 1

Activity	Student	Student's Information DB
Insert UTMID and Password	C	
Verify UTMID and Password		R

Process 2

Activity	Student	Admin	Shuttle Schedules DB
Schedules Request	C		R
Check Availability		RU	U
Display Shuttle Schedule	R		

Process 3

Activity	Student	Admin	Shuttle Routes DB
Search Shuttle's Number	C		
Track Shuttle's Location		RU	RU
Display Shuttle Live Location	R		

Process 4

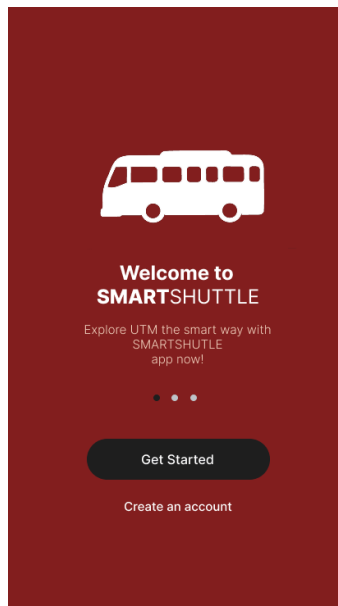
Activity	Student	Chatbot Logs
Enter Questions	C	R
AI Chatbot Answer	R	

Process 5

Activity	Student	Admin	Feedback DB	Report DB
Feedback Form	C	R	R	
Issue Report	C	R		R
Display Resolution	R	U		

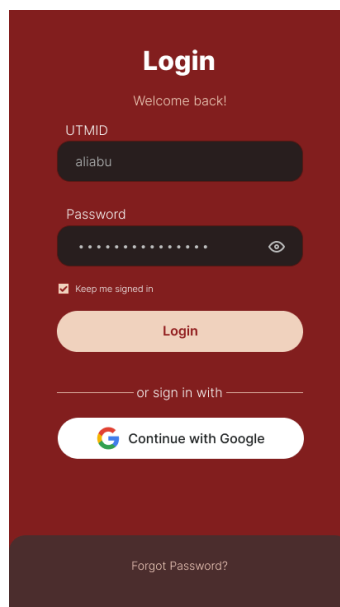
8.0 System Wireframe (Input Design, Output Design)

1. Welcome page



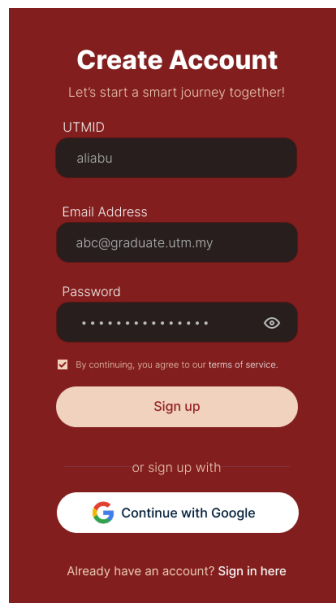
The welcome page in Smart Shuttle displays a friendly introduction screen with two main options: "Get Started" to login and "Create Account" for new users to sign up.

2. Login page



The login page allows users who already have an account to enter their UTM ID and password or sign in using their Google account for quick access.

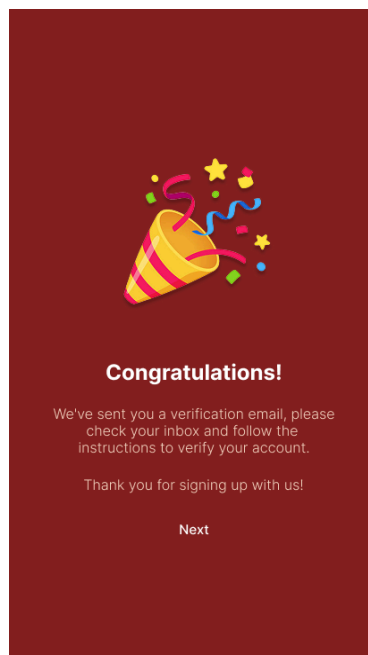
3. Account Register



The image shows a mobile app registration screen with a dark red background. At the top, the title "Create Account" is in white, followed by the subtitle "Let's start a smart journey together!". Below this are three input fields: "UTM ID" with the value "alliabu", "Email Address" with the value "abc@graduate.utm.my", and "Password" with masked characters and an eye icon. A checkbox is checked with the text "By continuing, you agree to our terms of service.". Below the inputs is a large orange "Sign up" button. Underneath the button is a link "or sign up with" followed by a "Continue with Google" button featuring the Google logo. At the bottom, there is a link "Already have an account? Sign in here".

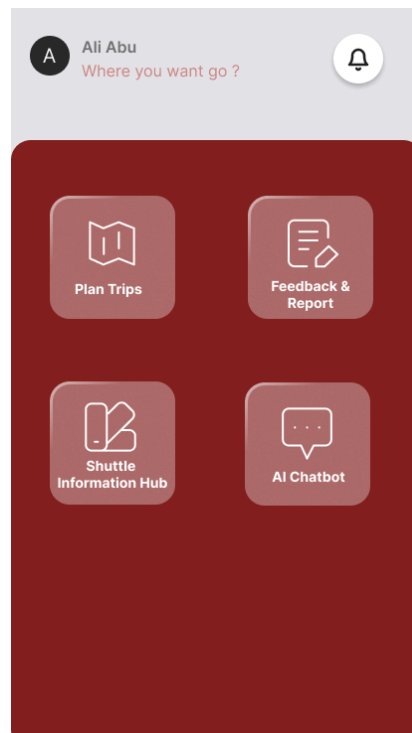
The account registration page requires users to fill in their UTM ID, email address, and password, then tap "Sign Up" to create a new account.

4. Signup Success Message



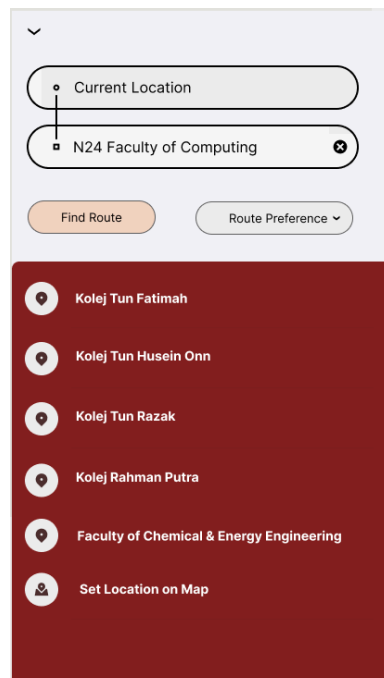
After successful signup, a message appears saying "Congratulations!" and informs the user that a verification email has been sent to their registered address.

5. Menu page



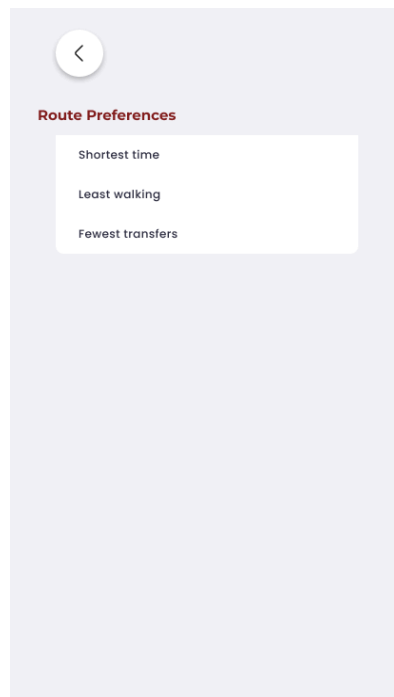
The menu page displays four main options: Plan Trip, Feedback & Report, Shuttle Information Hub, and AI Chat Box for easy navigation and access to key features.

6. Search Bar



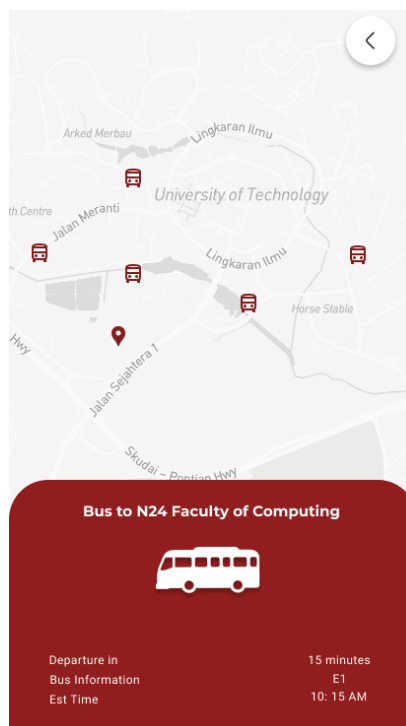
The search bar allows users to choose their location and also provides a list of suggested locations for quicker selection.

7. Smart Route Planner



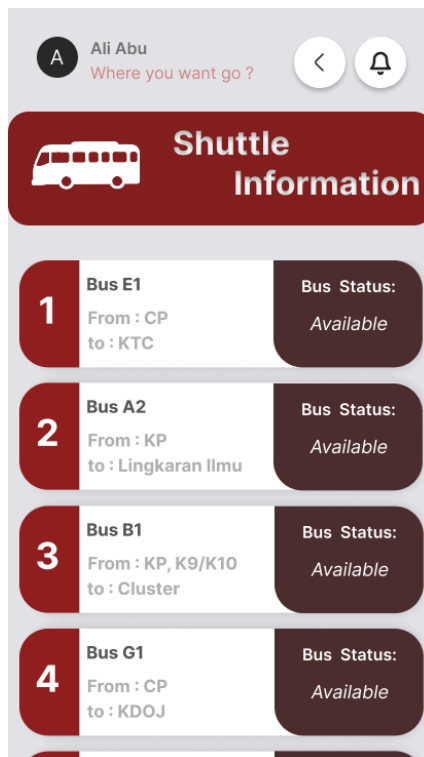
The Smart Route Planner lets users choose their route preference, such as the shortest time, least walking and fewer transfers route for a more personalized journey.

8. GPS Bus Tracking



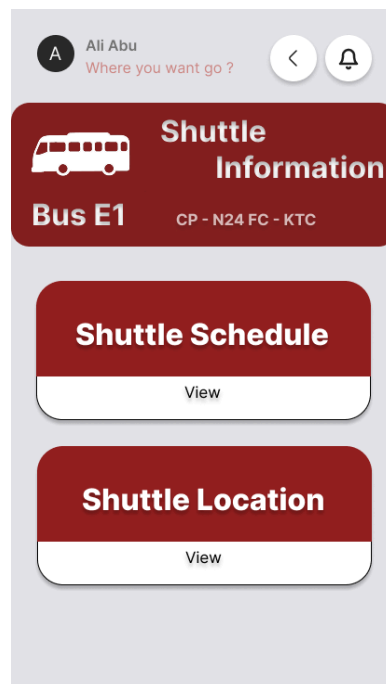
The GPS Bus Tracking feature shows the live location of the bus, along with its departure time, bus information, and estimated time of arrival (ETA).

9. Shuttle Information Hub



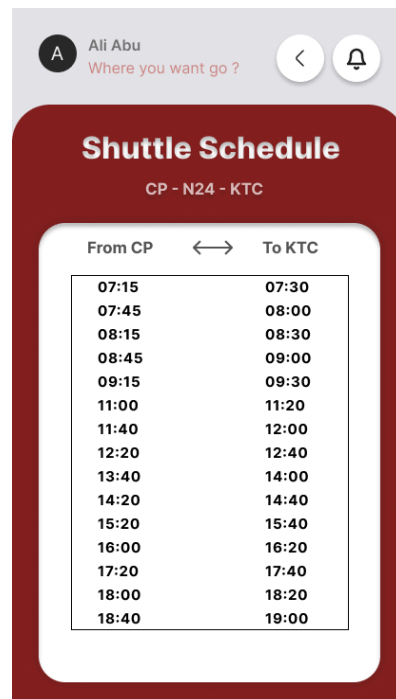
The Shuttle Information Hub provides a list of available buses, including the bus details.

10. Bus Details



When a bus is clicked on the previous page, it allows users to click a button to view the shuttle schedule and current location.

11. Shuttle Schedule

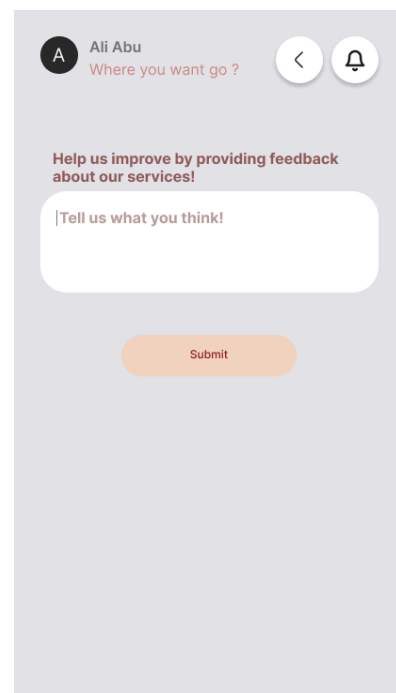


The image shows a mobile app interface for a shuttle schedule. At the top, there's a header with a user profile icon (A) and name 'Ali Abu', a red text prompt 'Where you want go?', and navigation icons (back and bell). Below this is a red card titled 'Shuttle Schedule' with the subtitle 'CP - N24 - KTC'. Inside the card is a white box containing a table with two columns: 'From CP' and 'To KTC', separated by a double-headed arrow. The table lists 16 departure times for each direction.

From CP	↔	To KTC
07:15		07:30
07:45		08:00
08:15		08:30
08:45		09:00
09:15		09:30
11:00		11:20
11:40		12:00
12:20		12:40
13:40		14:00
14:20		14:40
15:20		15:40
16:00		16:20
17:20		17:40
18:00		18:20
18:40		19:00

Users can view the shuttle schedule for better trip planning.

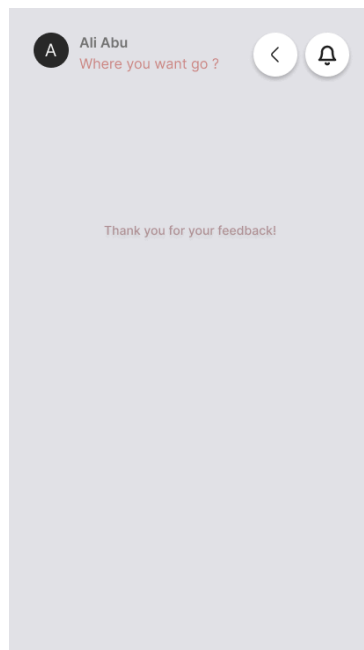
12. Feedback & Report



The image shows a mobile app interface for a feedback and report section. At the top, there's a header with a user profile icon (A) and name 'Ali Abu', a red text prompt 'Where you want go?', and navigation icons (back and bell). Below this is a light gray card with the text 'Help us improve by providing feedback about our services!'. Underneath is a white text input field with the placeholder text 'Tell us what you think!'. At the bottom of the card is an orange 'Submit' button.

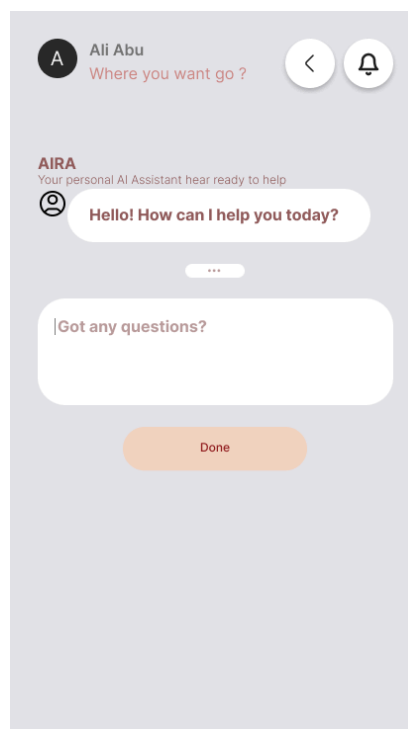
In the Feedback & Report section, users can fill in their feedback or report issues related to shuttle services, routes, or app functionality.

13. Submit Feedback & Report



After submitting, the page displays a confirmation message saying "Feedback Submitted" to inform the user that their input has been successfully received.

14. AI Chatbot



The AI Chatbot page provides users with a smart assistant to answer questions with any app-related inquiries.

9.0 Summary of the proposed system.

The UTM Smart Shuttle is an innovative transportation solution developed to overcome the inefficiencies of Universiti Teknologi Malaysia's existing campus shuttle system. As student populations grow and mobility demands increase, traditional methods of managing shuttle schedules and communication have become outdated, often leading to delays, confusion, and frustration among students especially. Recognizing these challenges, the UTM Smart Shuttle system was conceived as a comprehensive and intelligent platform designed to transform the campus free transportation experience.

UTM Smart Shuttle integrates improved features into a single, user-friendly mobile application. Among these features are real time shuttle tracking, which allows students to view the live location of buses, dynamic scheduling, which adjusts routes and timetables based on demand and automated notifications, which keep users informed about delays or route changes. The system also incorporates an AI-powered chatbot that handles student inquiries, providing instant answers to common questions about shuttle bus services. Additionally, a feedback and reporting mechanism is included, empowering students to report issues or suggest improvements directly through the app.

By consolidating these components, the Smart Shuttle system significantly enhances both the student experience and UTM Fleet administrative efficiency. Students benefit from greater transparency, as they no longer need to guess shuttle arrival times or rely on outdated information. The application's automated features reduce the need for manual intervention, allowing the UTM Fleet Administration to focus on optimizing fleet performance and responding more effectively to operational issues.

Beyond solving current transportation problems, the UTM Smart Shuttle represents a major step forward in the university's journey toward becoming a smart campus. By leveraging real time data, automation, and user engagement, the system demonstrates how digital transformation can be applied to public services within an academic environment. It creates a more connected, efficient, and responsive transportation ecosystem, which not only meets today's needs but also lays the foundation for future innovations in campus mobility.

In conclusion, the UTM Smart Shuttle is more than just a mobile app, it is a strategic solution that redefines how campus transportation is managed and experienced. Through its thoughtful integration of technology, it brings measurable improvements in efficiency, reliability, and user satisfaction, benefitting both students and the UTM Fleet administrators alike.